2.2 Calculus of Vector-Valued Functions

1. Determine if the following limits exist. If it does not exist, you must explain why.

a) If $\mathbf{r}(t)=\dfrac{t-2}{t^2-4}\mathbf{i}+\dfrac{\cos(\pi/t)}{t-2}\mathbf{j}-\tan^{-1}\left(\dfrac{1}{|t-2|}\right)\mathbf{k}$, then find $\lim\limits_{t\to 2}\mathbf{r}(t)$.

b) If $\mathbf{r}(t)=e^{-3t}\mathbf{i}+\dfrac{t^2}{\sin^2(t)}\mathbf{j}+\cos(2t)\mathbf{k}$, then find $\lim\limits_{t\to 0}\mathbf{r}(t)$.

c) Find $\lim\limits_{t\to\infty}\left\langle \dfrac{1+t^2}{1-t^2}, \tan^{-1}(t), \dfrac{1-e^{-2t}}{t} \right\rangle$.

d) Find $\lim_{t\to\infty}\left\langle \frac{\ln(t)}{2\sqrt{t}}, te^{-t}, \frac{2t^5-5t}{1+t^5} \right\rangle$.

2. Find the parametric equation for the tangent line to the curve with given parametric equations at the specified point.

$$x = 2+\ln(t),\quad y = 3t^3,\quad z = 3t+1;\quad (2,3,4)$$

3. Let $\mathbf{r}(t)=\langle 5\tan^{-1}(t-1),\sinh(t-2),-\ln(t-1)\rangle$. Write parametric equations of the tangent line to the graph of $\mathbf{r}(t)$ at $\left(\frac{5\pi}{4},0,0\right)$.

4. Find a vector function $\mathbf{L}(t)$ which is the tangent line to the following curves at the following points.

a) $\mathbf{r}(t)=\langle 1+2\sqrt{t},t^3-t,t^3+t\rangle$ at the point $(3,0,2)$

b) $\mathbf{r}(t)=\left\langle e^t,te^t,te^{t^2}\right\rangle$ at the point $(1,0,0)$

c) $\mathbf{r}(t)=\left\langle e^{-t}\cos(t), e^{-t}\sin(t), e^{-t}\right\rangle$ at the point $(1,0,1)$

d) $\mathbf{r}(t)=\left\langle \sqrt{t^2+3}, \ln\left(t^2+3\right), t\right\rangle$ at the point $(2,\ln(4),1)$

5. Let $\mathbf{r}(t)=\left\langle e^{2t}\cos(2t), 5, e^{2t}\sin(2t)\right\rangle$, find $\left\|\mathbf{r}'(t)\right\|$. Fully simplify the final solution.

6. Two particles travel along the space curves

$$\mathbf{r}_1(t) = \langle t, t^2, t^3 \rangle, \qquad \mathbf{r}_2(t) = \langle 1+2t, 1+6t, 1+14t \rangle$$

Do the particles collide? Do their paths intersect?

7. The curves $\mathbf{r}_1(s) = \left\langle 3(s-1), 2(s-1), 5(s-1)^2 \right\rangle$ and $\mathbf{r}_2(t) = \langle \sin(t), \sin(2t), t \rangle$ intersect at the origin. Find their angle of intersection.

8. Evaluate each of the following integrals.

a) $\displaystyle\int_0^2 \left(t\mathbf{i} - t^3\mathbf{j} + 3t^5\mathbf{k}\right) dt$

b) $\displaystyle\int_0^1 \left(\frac{4}{1+t^2}\mathbf{j} + \frac{2t}{1+t^2}\mathbf{k}\right) dt$

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c) $\displaystyle\int_0^{\pi/2} \left\langle 3\sin^2(t)\cos(t), 3\sin(t)\cos^2(t), 2\sin(t)\cos(t)\right\rangle dt$

d) $\int_1^2 \left\langle t^2, t\sqrt{t-1}, t\sin(\pi t) \right\rangle dt$

e) $\int \left(\sec^2(t)\,\mathbf{i} + t\left(t^2+1\right)^3 \mathbf{j} + t^2 \ln(t)\,\mathbf{k} \right) dt$.

f) $\int \left\langle te^{2t}, \frac{t}{1-t}, \frac{1}{\sqrt{1-t^2}} \right\rangle dt$

9. Let $\mathbf{r}(t) = \left\langle 2, \sqrt{1+t}, \frac{2}{t^2-3t-4} \right\rangle$. Evaluate $\int_0^3 \mathbf{r}(t)\,dt$.

10. Determine the function $\mathbf{r}(t)$ which satisfies each of the given conditions.

a) $\mathbf{r}'(t) = \langle 2t, 3t^2, 1 \rangle$ and $\mathbf{r}(0) = \langle 1, -2, 4 \rangle$.

b) $\mathbf{r}'(t) = \langle \cos(t), 2\sin(t), \sec^2(t) \rangle$ and $\mathbf{r}(\pi) = \langle 3, 1, -1 \rangle$.

c) $\mathbf{r}''(t)=\langle 6t, e^{t}, -\cos(t)\rangle$ and $\mathbf{r}'(0)=\langle 0,1,2\rangle$ & $\mathbf{r}(0)=\langle 1,0,-1\rangle$.

d) $\mathbf{r}'(t)=\left\langle e^{2t}, \dfrac{1}{t+1}, te^{t^2}\right\rangle$ and $\mathbf{r}(0)=\langle 0,2,-3\rangle$.

11. Let $\mathbf{r}'(t)=\frac{1}{1+t^2}\mathbf{i}-\frac{1}{\sqrt{1-t^2}}\mathbf{j}+\frac{t}{t^2+1}\mathbf{k}$ and $\mathbf{r}(0)=\mathbf{i}+\frac{\pi}{2}\mathbf{j}+2\mathbf{k}$. Determine $\mathbf{r}(t)$.

12. Suppose that we have vector functions **u** and **v** where $\mathbf{u}(2)=\langle 1,2,-1\rangle$, $\mathbf{u}'(2)=\langle 3,0,4\rangle$, and $\mathbf{v}(t)=\langle t,t^2,t^3\rangle$.

a) Find $f'(2)$, where $f(t)=\mathbf{u}(t)\cdot\mathbf{v}(t)$.

b) Find $\mathbf{r}'(2)$, where $\mathbf{r}(t)=\mathbf{u}(t)\times\mathbf{v}(t)$.

13. Show that if $\mathbf{r}$ is a vector function such that $\mathbf{r}''$ exists, then $\frac{d}{dt}\left[\mathbf{r}(t)\times\mathbf{r}'(t)\right]=\mathbf{r}(t)\times\mathbf{r}''(t)$.

14. Find an expression for $\frac{d}{dt}\left[\mathbf{u}(t)\cdot\left(\mathbf{v}(t)\times\mathbf{w}(t)\right)\right]$.